

maximoN

maximo

```
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.4.3

Warning: package 'lubridate' was built under R version 4.4.3

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.1.4      v readr      2.1.5
```

```
v forcats   1.0.0      v stringr    1.5.1
```

```
v ggplot2    3.5.2      v tibble     3.2.1
```

```
v lubridate  1.9.4      v tidyr      1.3.1
```

```
v purrr      1.0.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(readxl)
```

Warning: package 'readxl' was built under R version 4.4.2

```
library(lme4)
```

Warning: package 'lme4' was built under R version 4.4.2

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

```
library(skimr)
```

Warning: package 'skimr' was built under R version 4.4.2

```
library(emmeans)
```

Warning: package 'emmeans' was built under R version 4.4.2

Welcome to emmeans.

Caution: You lose important information if you filter this package's results.

See '? untidy'

```
library(nlme)
```

Attaching package: 'nlme'

The following object is masked from 'package:lme4':

lmList

The following object is masked from 'package:dplyr':

collapse

```
library(nlraa)
```

For points

```
variety.testing <- read.csv("2025-maximo-and-elvis/raw-data/Corn data organized-10-16-2024.csv")
variety.testing$yield <- as.numeric(variety.testing$yield)
```

Warning: NAs introduced by coercion

```
skim(variety.testing)
```

Table 1: Data summary

Name	variety.testing
Number of rows	4759
Number of columns	24
Column type frequency:	
character	6
numeric	18
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Location	0	1	6	10	0	7	0
Hybrid	0	1	6	32	0	767	0
Yield.bu.acre	0	1	3	6	0	1159	0
soil_textural_class	0	1	12	23	0	8	0
sept	0	1	0	6	275	25	0
week	0	1	9	11	0	5	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
Year	0	1.00	2015.59	3.91	2010.00	2012.00	2015.00	2019.00	2023.00	
SowingdateDOY	0	1.00	106.90	10.57	87.00	98.00	107.00	112.00	136.00	
HarvestdateDOY	0	1.00	250.75	11.52	227.00	244.00	251.00	256.00	283.00	
CRM	240	0.95	115.21	2.28	103.00	114.00	115.00	117.00	121.00	
yield	57	0.99	11748.33	1406.37	6645.00	10809.40	11680.25	12739.30	16803.80	
mean_maxt	114	0.98	30.96	1.23	28.73	29.96	31.05	31.75	34.90	
mean_mint	114	0.98	19.92	0.88	18.09	19.17	19.99	20.60	22.80	
cum_radn	114	0.98	3196.99	164.56	2418.30	3106.78	3196.89	3308.98	3618.42	
cum_rain	114	0.98	459.68	111.58	211.98	400.80	483.60	531.93	678.64	
avg_t	114	0.98	25.44	0.98	23.46	24.63	25.43	26.12	28.80	
above_32	114	0.98	65.62	16.56	32.00	51.00	68.00	79.00	97.00	
row_spacing	0	1.00	0.87	0.10	0.76	0.76	0.97	0.97	0.97	

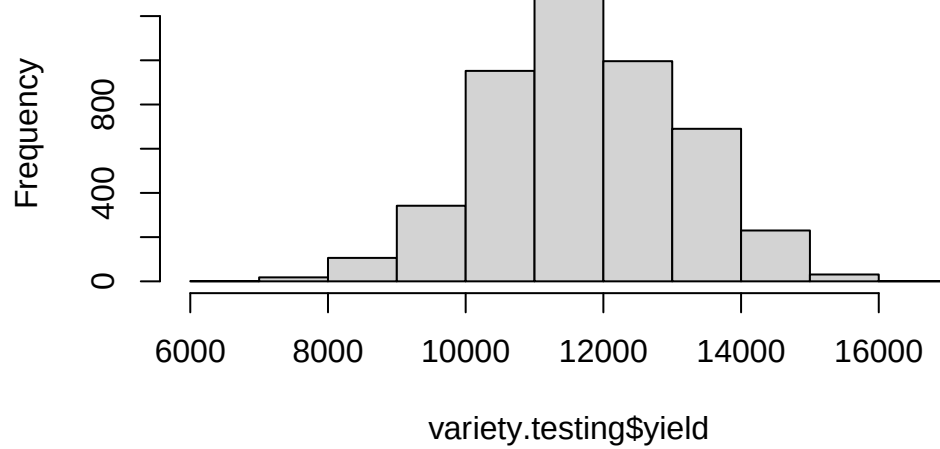
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
april	275	0.94	156.36	66.01	28.40	116.84	155.40	187.96	345.20	
may	275	0.94	133.58	68.29	12.70	91.44	137.16	175.26	295.40	
june	275	0.94	92.48	73.44	19.80	53.34	76.20	114.30	510.54	
july	275	0.94	95.90	58.80	17.78	60.96	81.28	111.30	286.50	
august	275	0.94	88.57	59.45	0.00	38.10	76.20	137.16	208.28	
Precipitation	275	0.94	588.35	178.29	154.94	449.58	604.52	680.72	1023.62	

```
str(variety.testing)
```

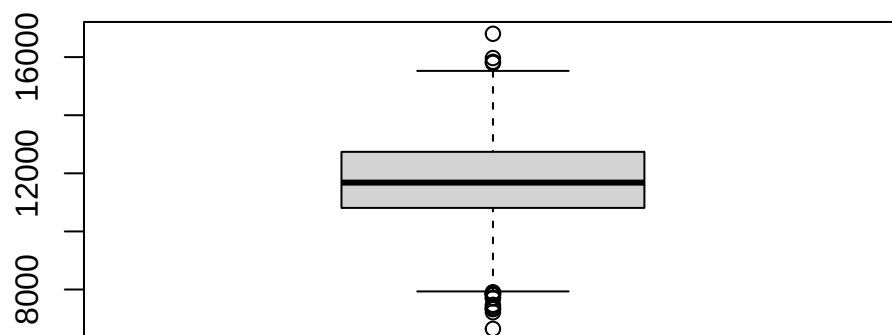
```
'data.frame':  4759 obs. of  24 variables:
 $ Location      : chr  "Keiser" "Keiser" "Keiser" "Keiser" ...
 $ Year          : int   2010 2010 2010 2010 2010 2010 2010 2010 2010 2010 2010 ...
 $ Hybrid        : chr   "M-Pride MP3150" "DEKALB DKC64-69 (GENVT3P)" "Delta Grow 3788" ...
 $ Yield.bu.acre : chr   "252.5" "234.5" "233" "229.6" ...
 $ SowingdateDOY : int   108 108 108 108 108 108 108 108 108 108 ...
 $ HarvestdateDOY : int   244 244 244 244 244 244 244 244 244 244 ...
 $ CRM           : int    NA 114 115 116 NA 115 NA NA 116 116 ...
 $ yield         : num  13498 12536 12456 12274 12237 ...
 $ mean_maxt     : num   31 31 31 31 31 31 31 31 31 31 ...
 $ mean_mint     : num   21 21 21 21 21 21 21 21 21 21 ...
 $ cum_radn      : num  3161 3161 3161 3161 3161 ...
 $ cum_rain      : num   480 480 480 480 480 480 480 480 480 480 ...
 $ avg_t         : num   26 26 26 26 26 26 26 26 26 26 ...
 $ above_32      : int   71 71 71 71 71 71 71 71 71 71 ...
 $ soil_textural_class: chr   "sharkey clay" "sharkey clay" "sharkey clay" "sharkey clay" ...
 $ row_spacing   : num   0.965 0.965 0.965 0.965 0.965 ...
 $ april         : num   68.6 68.6 68.6 68.6 68.6 68.6 68.6 68.6 68.6 68.6 ...
 $ may           : num  238 238 238 238 238 ...
 $ june          : num   24.1 24.1 24.1 24.1 24.1 24.1 24.1 24.1 24.1 24.1 ...
 $ july          : num   89.9 89.9 89.9 89.9 89.9 89.9 89.9 89.9 89.9 89.9 ...
 $ august        : num    1.3 1.3 1.3 1.3 1.3 1.3 1.3 1.3 1.3 1.3 ...
 $ sept          : chr    "0" "0" "0" "0" ...
 $ Precipitation : num  422 422 422 422 422 ...
 $ week          : chr   "April 11-20" "April 11-20" "April 11-20" "April 11-20" ...
```

```
hist(variety.testing$yield)
```

Histogram of variety.testing\$yield



```
boxplot(variety.testing$yield)
```



```

variety.testing <- variety.testing %>%
  mutate(CRM_group = case_when(
    CRM %in% c(103, 107, 108) ~ "Group_1",
    CRM %in% c(110, 111, 112, 113) ~ "Group_2",
    CRM %in% c(114, 115, 116, 117) ~ "Group_3",
    CRM %in% c(118, 119, 120, 121) ~ "Group_4"
  ))
variety.testing <- variety.testing %>%
  mutate(
    Year = as.factor(Year),
    Location = as.factor(Location),
    CRM_group = as.factor(CRM_group),
    yield = as.numeric(yield)
  )
variety.testing <- na.omit(variety.testing)
str(variety.testing)

```

'data.frame': 4073 obs. of 25 variables:

```

$ Location      : Factor w/ 7 levels "Bellfarm","Greenfield",...: 4 4 4 4 4 4 4 4 4 4 ...
$ Year          : Factor w/ 14 levels "2010","2011",...: 1 1 1 1 1 1 1 1 1 1 ...
$ Hybrid        : chr  "DEKALB DKC64-69 (GENVT3P)" "Delta Grow 3788" "REV26HR50" "REV26HR50" ...
$ Yield.bu.acre : chr  "234.5" "233" "229.6" "225.1" ...
$ SowingdateDOY : int  108 108 108 108 108 108 108 108 108 108 ...
$ HarvestdateDOY : int  244 244 244 244 244 244 244 244 244 244 ...
$ CRM           : int  114 115 116 115 116 116 116 113 115 113 ...
$ yield         : num  12536 12456 12274 12034 11670 ...
$ mean_maxt     : num  31 31 31 31 31 31 31 31 31 31 ...
$ mean_mint     : num  21 21 21 21 21 21 21 21 21 21 ...
$ cum_radn      : num  3161 3161 3161 3161 3161 ...
$ cum_rain      : num  480 480 480 480 480 480 480 480 480 480 ...
$ avg_t         : num  26 26 26 26 26 26 26 26 26 26 ...
$ above_32      : int  71 71 71 71 71 71 71 71 71 71 ...
$ soil_textural_class: chr  "sharkey clay" "sharkey clay" "sharkey clay" "sharkey clay" ...
$ row_spacing   : num  0.965 0.965 0.965 0.965 0.965 ...
$ april         : num  68.6 68.6 68.6 68.6 68.6 68.6 68.6 68.6 68.6 68.6 ...
$ may           : num  238 238 238 238 238 ...
$ june          : num  24.1 24.1 24.1 24.1 24.1 24.1 24.1 24.1 24.1 24.1 ...
$ july          : num  89.9 89.9 89.9 89.9 89.9 89.9 89.9 89.9 89.9 89.9 ...
$ august        : num  1.3 1.3 1.3 1.3 1.3 1.3 1.3 1.3 1.3 1.3 ...
$ sept          : chr  "0" "0" "0" "0" ...
$ Precipitation : num  422 422 422 422 422 ...
$ week          : chr  "April 11-20" "April 11-20" "April 11-20" "April 11-20" ...

```

```
$ CRM_group          : Factor w/ 4 levels "Group_1","Group_2",...: 3 3 3 3 3 3 3 2 3 2 ...
- attr(*, "na.action")= 'omit' Named int [1:686] 1 5 7 8 13 14 15 16 17 19 ...
..- attr(*, "names")= chr [1:686] "1" "5" "7" "8" ...
```

###Fitting models

```
str(variety.testing)
```

```
'data.frame': 4073 obs. of 25 variables:
 $ Location          : Factor w/ 7 levels "Bellfarm","Greenfield",...: 4 4 4 4 4 4 4 4 4 4 .
 $ Year              : Factor w/ 14 levels "2010","2011",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ Hybrid            : chr "DEKALB DKC64-69 (GENVT3P)" "Delta Grow 3788" "REV26HR50" "REV26HR50" ...
 $ Yield.bu.acre     : chr "234.5" "233" "229.6" "225.1" ...
 $ SowingdateDOY     : int 108 108 108 108 108 108 108 108 108 108 ...
 $ HarvestdateDOY    : int 244 244 244 244 244 244 244 244 244 244 ...
 $ CRM               : int 114 115 116 115 116 116 116 113 115 113 ...
 $ yield             : num 12536 12456 12274 12034 11670 ...
 $ mean_maxt         : num 31 31 31 31 31 31 31 31 31 31 ...
 $ mean_mint         : num 21 21 21 21 21 21 21 21 21 21 ...
 $ cum_radn          : num 3161 3161 3161 3161 3161 ...
 $ cum_rain          : num 480 480 480 480 480 480 480 480 480 480 ...
 $ avg_t             : num 26 26 26 26 26 26 26 26 26 26 ...
 $ above_32          : int 71 71 71 71 71 71 71 71 71 71 ...
 $ soil_textural_class: chr "sharkey clay" "sharkey clay" "sharkey clay" "sharkey clay" ...
 $ row_spacing       : num 0.965 0.965 0.965 0.965 0.965 ...
 $ april             : num 68.6 68.6 68.6 68.6 68.6 68.6 68.6 68.6 68.6 68.6 ...
 $ may               : num 238 238 238 238 238 ...
 $ june              : num 24.1 24.1 24.1 24.1 24.1 24.1 24.1 24.1 24.1 24.1 ...
 $ july              : num 89.9 89.9 89.9 89.9 89.9 89.9 89.9 89.9 89.9 89.9 ...
 $ august            : num 1.3 1.3 1.3 1.3 1.3 1.3 1.3 1.3 1.3 1.3 ...
 $ sept              : chr "0" "0" "0" "0" ...
 $ Precipitation     : num 422 422 422 422 422 ...
 $ week              : chr "April 11-20" "April 11-20" "April 11-20" "April 11-20" ...
 $ CRM_group         : Factor w/ 4 levels "Group_1","Group_2",...: 3 3 3 3 3 3 3 2 3 2 ...
- attr(*, "na.action")= 'omit' Named int [1:686] 1 5 7 8 13 14 15 16 17 19 ...
..- attr(*, "names")= chr [1:686] "1" "5" "7" "8" ...
```

```
mod1 <- lme(yield ~ CRM_group * SowingdateDOY,
            random= ~1| Location,
            data= variety.testing)
anova(mod1)
```

	numDF	denDF	F-value	p-value
(Intercept)	1	4061	17868.331	<.0001
CRM_group	3	4061	2.219	0.0838
SowingdateDOY	1	4061	18.665	<.0001
CRM_group:SowingdateDOY	3	4061	1.775	0.1497

Interaction between CRM and Sowing date is not significant. Remove the interaction to simplify the model.

Mod2

```
mod2 <- lme(yield ~ CRM_group * SowingdateDOY,
            random= ~1|Location,
            data= variety.testing)
anova(mod2)
```

	numDF	denDF	F-value	p-value
(Intercept)	1	4061	17868.331	<.0001
CRM_group	3	4061	2.219	0.0838
SowingdateDOY	1	4061	18.665	<.0001
CRM_group:SowingdateDOY	3	4061	1.775	0.1497

```
IC_tab(mod1,mod2)
```

	model	df	AIC	AICc	BIC	dAIC	weight
1	mod1	10	70242.49	70242.55	70305.59	0	0.5
2	mod2	10	70242.49	70242.55	70305.59	0	0.5

Even though `crm * sowing date` is not significant, the model with the interaction is better as it has a lower AIC.

Adding Year and hybrid as random effects. When adding Year as a random effect, sowing date is no longer significant. Probably because there is 1 sowing date per year/location combination.

```
mod2.1 <- lme(yield ~ CRM_group * SowingdateDOY,
              random = list(Location = ~1, Year = ~1),
              data = variety.testing)
anova(mod2.1)
```

	numDF	denDF	F-value	p-value
(Intercept)	1	4012	5952.761	<.0001
CRM_group	3	4012	34.517	<.0001
SowingdateDOY	1	49	1.662	0.2034
CRM_group:SowingdateDOY	3	4012	0.269	0.8478

```
IC_tab(mod2, mod2.1)
```

	model	df	AIC	AICc	BIC	dAIC	weight
2	mod2.1	11	66303.98	66304.04	66373.39	0.000	1
1	mod2	10	70242.49	70242.55	70305.59	3938.513	0

Adding hybrid as a random effect

```
mod2.2 <- lme(yield ~ CRM_group * SowingdateDOY,
              random = list(Location = ~1, Year = ~1, Hybrid = ~1),
              data = variety.testing)
anova(mod2.2)
```

	numDF	denDF	F-value	p-value
(Intercept)	1	4012	5954.343	<.0001
CRM_group	3	4012	34.517	<.0001
SowingdateDOY	1	49	1.663	0.2033
CRM_group:SowingdateDOY	3	4012	0.269	0.8478

```
IC_tab(mod2.1, mod2.2)
```

	model	df	AIC	AICc	BIC	dAIC	weight
1	mod2.1	11	66303.98	66304.04	66373.39	0.000000	0.7310599
2	mod2.2	12	66305.98	66306.06	66381.70	2.000013	0.2689401

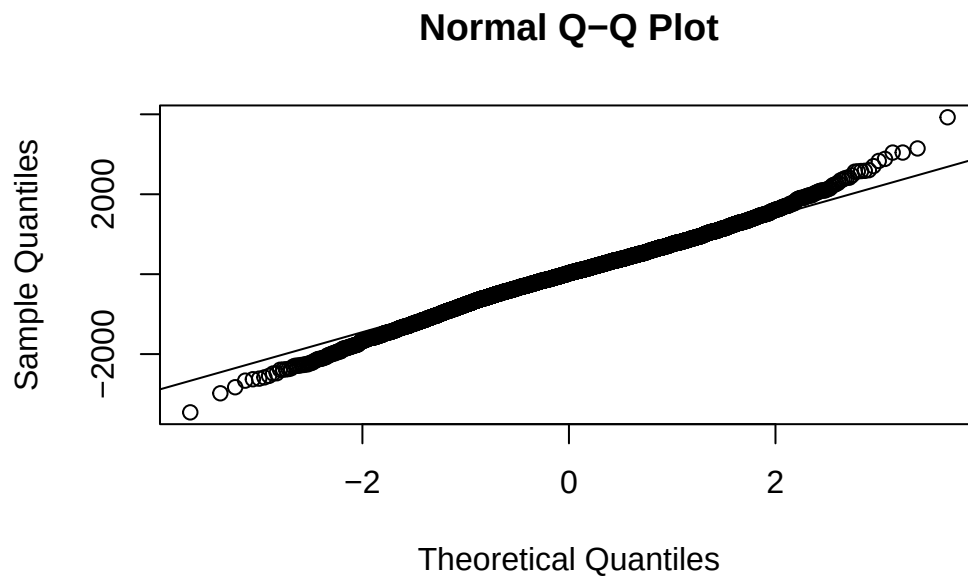
Adding hybrid did not improve the model. Now lets test if nesting location within year improves the model.

```
mod2.3 <- lme(yield ~ CRM_group * SowingdateDOY,
              random = ~1 | Year/Location,
              data = variety.testing)
IC_tab(mod2.1, mod2.3)
```

	model	df	AIC	AICc	BIC	dAIC	weight
2	mod2.3	11	66295.16	66295.22	66364.57	0.000000	0.98800746
1	mod2.1	11	66303.98	66304.04	66373.39	8.822811	0.01199254

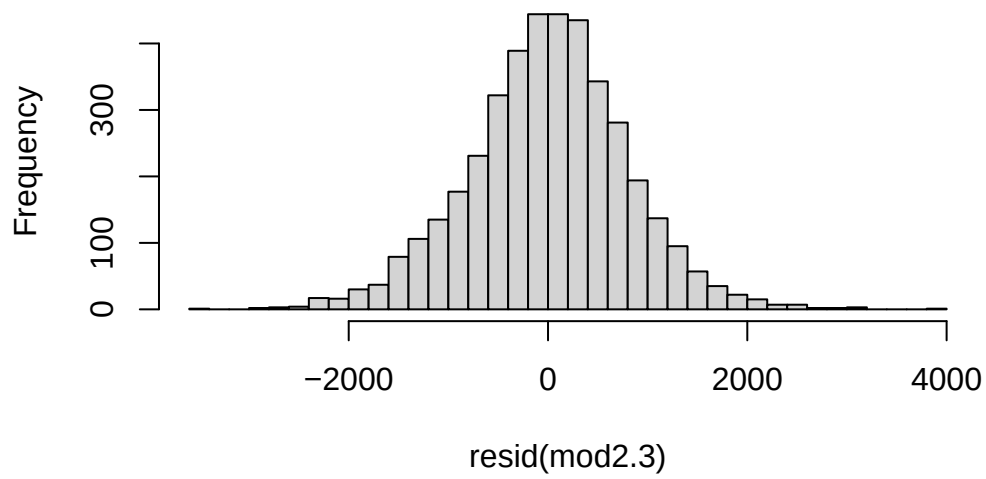
Nesting Location in year improved the model, lets check residuals.

```
qqnorm(resid(mod2.3))
qqline(resid(mod2.3))
```

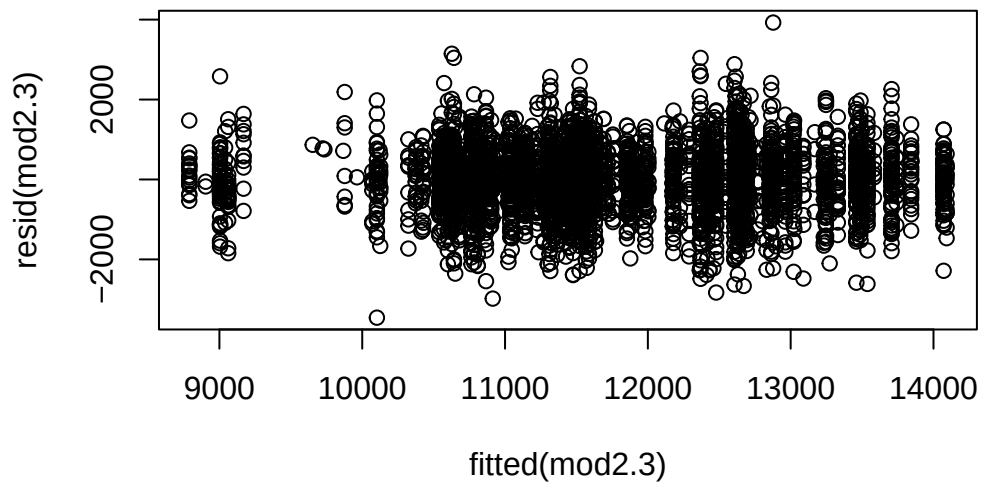


```
hist(resid(mod2.3), breaks = 50)
```

Histogram of resid(mod2.3)



```
plot(fitted(mod2.3), resid(mod2.3))
```



```
shapiro.test(resid(mod2.3))
```

Shapiro-Wilk normality test

```
data:  resid(mod2.3)
W = 0.9952, p-value = 2.635e-10
```

Residuals look homogeneously distributed and fairly normal. However, Shapiro test indicates residuals are not normally distributed. Now modify var co-var structure to improve model fit.

Var-Covar

```
mod2_loc <- lme(yield ~ CRM_group * SowingdateDOY,
               random = ~1 | Year/Location,
               weights = varIdent(form= ~1 | Location),
               data= variety.testing)
anova(mod2_loc)
```

	numDF	denDF	F-value	p-value
(Intercept)	1	4012	2572.5535	<.0001
CRM_group	3	4012	34.5615	<.0001
SowingdateDOY	1	40	1.5817	0.2158
CRM_group:SowingdateDOY	3	4012	0.4246	0.7353

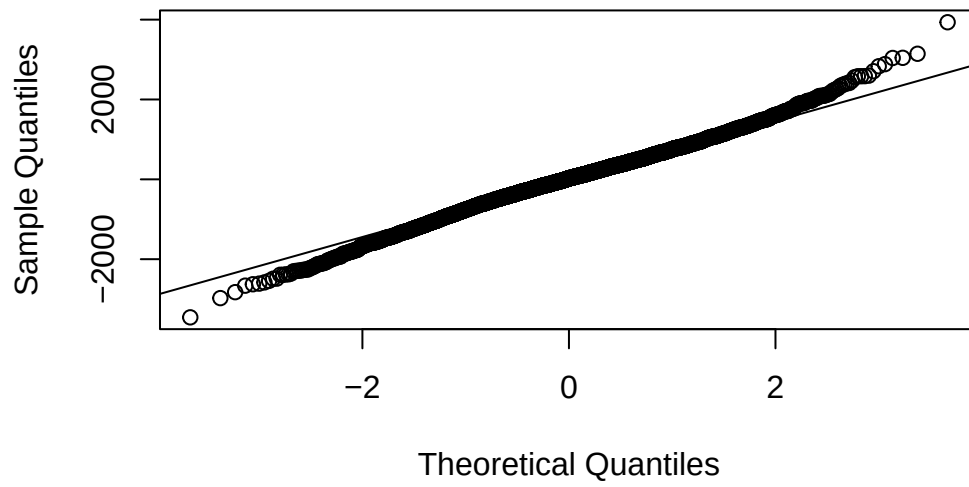
```
IC_tab(mod2.3, mod2_loc)
```

	model	df	AIC	AICc	BIC	dAIC	weight
2	mod2_loc	15	66278.41	66278.52	66373.06	0.00000	0.9997695452
1	mod2.3	11	66295.16	66295.22	66364.57	16.75045	0.0002304548

Covar structure improved the model Lets test residuals

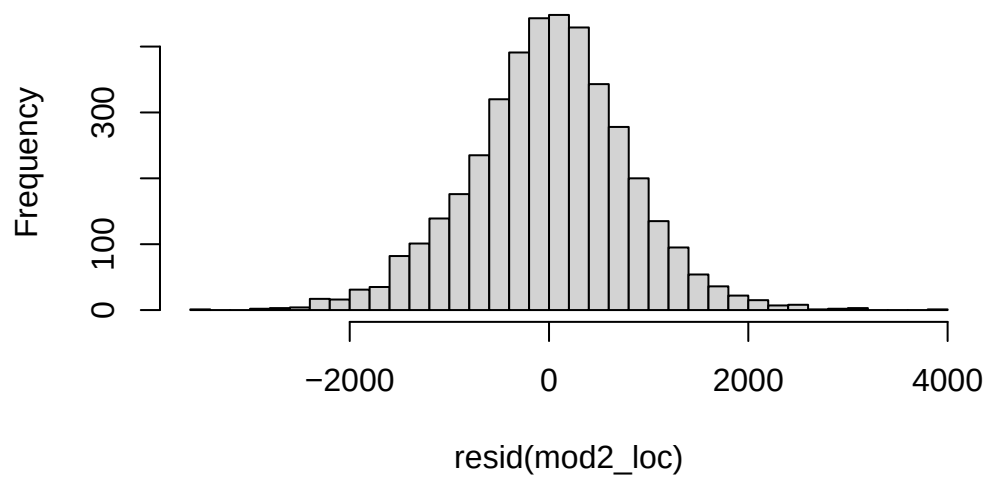
```
qqnorm(resid(mod2_loc))
qqline(resid(mod2_loc))
```

Normal Q-Q Plot

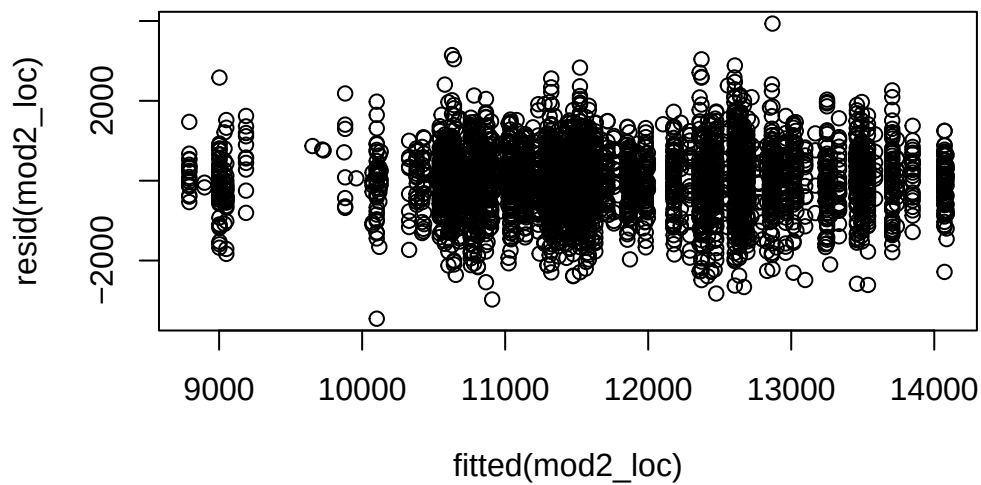


```
hist(resid(mod2_loc), breaks = 50)
```

Histogram of resid(mod2_loc)



```
plot(fitted(mod2_loc), resid(mod2_loc))
```



```
shapiro.test(resid(mod2_loc))
```

Shapiro-Wilk normality test

data: resid(mod2_loc)

W = 0.99518, p-value = 2.431e-10

#For points

```
variety.testing <- variety.testing %>%  
  mutate(CRM_group = case_when(  
    #CRM %in% c(103, 107, 108) ~ "Group_1",  
    CRM %in% c(110, 111, 112, 113) ~ "Group_2",  
    CRM %in% c(114, 115, 116, 117) ~ "Group_3",  
    CRM %in% c(118, 119, 120, 121) ~ "Group_4"  
  ))  
variety.testing <- variety.testing %>%  
  mutate(  
    #CRM %in% c(103, 107, 108) ~ "Group_1",  
    CRM %in% c(110, 111, 112, 113) ~ "Group_2",  
    CRM %in% c(114, 115, 116, 117) ~ "Group_3",  
    CRM %in% c(118, 119, 120, 121) ~ "Group_4"
```

```

    Year = as.factor(Year),
    Location = as.factor(Location),
    CRM_group = as.factor(CRM_group),
    yield = as.numeric(yield),
    SowingdateDOY = as.factor(SowingdateDOY)
  )
variety.testing <- na.omit(variety.testing)

```

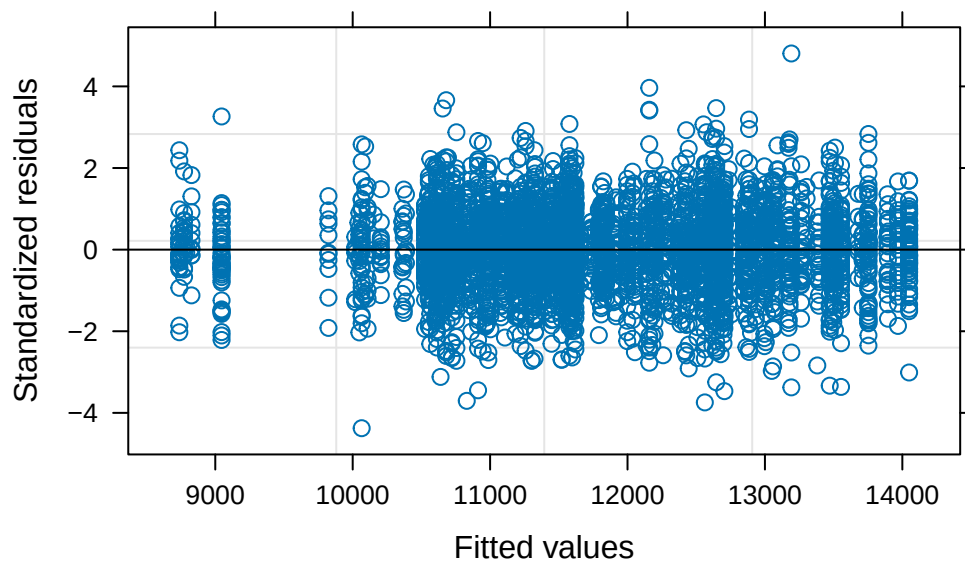
```

mod2_loc.a <- lme(yield ~ CRM_group * SowingdateDOY,
  random = ~1 | Year/Location,
  weights = varIdent(form= ~1 | Location),
  data= variety.testing)
emmeans.model.a <- emmeans(mod2_loc.a,"SowingdateDOY") %>%
  as.data.frame() %>%
  mutate(SowingdateDOY=as.numeric(as.character(SowingdateDOY)))

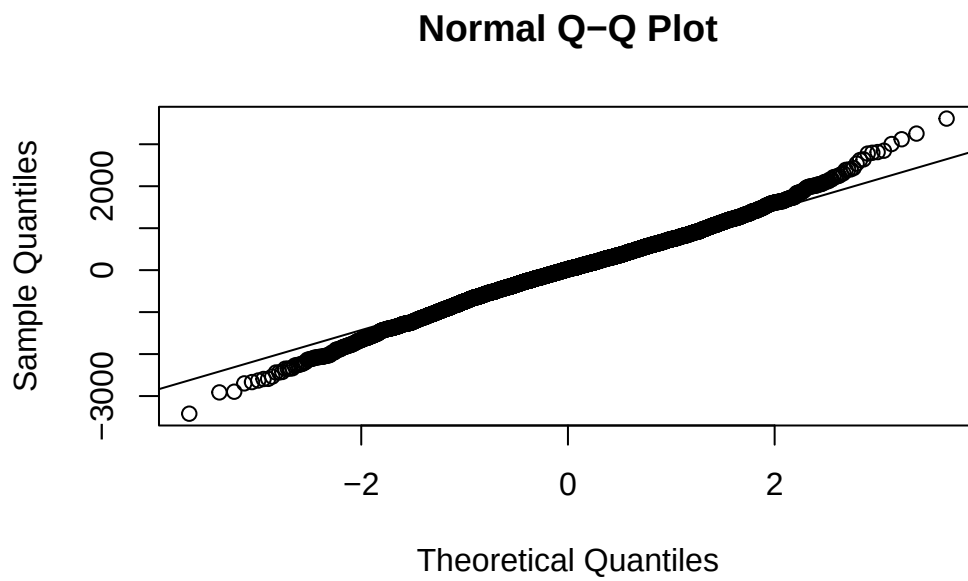
```

NOTE: Results may be misleading due to involvement in interactions

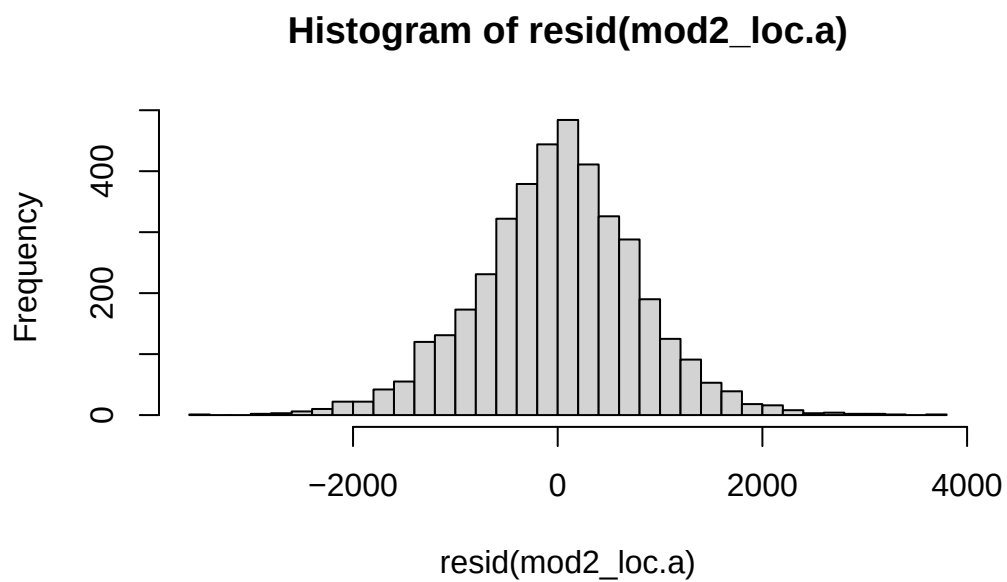
```
plot(mod2_loc.a)
```



```
qqnorm(resid(mod2_loc.a)); qqline(resid(mod2_loc.a))
```



```
hist(resid(mod2_loc.a), breaks = 30)
```



Lets get the overall slope of each effect. Check with interaction between sowing and without it by changing mod2_loc.

##For slope

```
variety.testing1 <- read.csv("2025-maximo-and-elvis/raw-data/Corn data organized-10-16-2024.csv")

variety.testing1 <- variety.testing1 %>%
  mutate(CRM_group = case_when(
    CRM %in% c(103, 107, 108) ~ "Group_1",
    CRM %in% c(110, 111, 112, 113) ~ "Group_2",
    CRM %in% c(114, 115, 116, 117) ~ "Group_3",
    CRM %in% c(118, 119, 120, 121) ~ "Group_4"
  ))
variety.testing1 <- variety.testing1 %>%
  mutate(
    Year = as.factor(Year),
    Location = as.factor(Location),
    CRM_group = as.factor(CRM_group),
    yield = as.numeric(yield),
  )
```

Warning: There was 1 warning in `mutate()`.
i In argument: `yield = as.numeric(yield)`.
Caused by warning:
! NAs introduced by coercion

```
variety.testing1 <- na.omit(variety.testing1)
str(variety.testing1)
```

```
'data.frame':  4073 obs. of  25 variables:
 $ Location      : Factor w/ 7 levels "Bellfarm","Greenfield",...: 4 4 4 4 4 4 4 4 4 4 ...
 $ Year          : Factor w/ 14 levels "2010","2011",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ Hybrid        : chr  "DEKALB DKC64-69 (GENVT3P)" "Delta Grow 3788" "REV26HR50" "REV26HR50" ...
 $ Yield.bu.acre : chr  "234.5" "233" "229.6" "225.1" ...
 $ SowingdateDOY : int   108 108 108 108 108 108 108 108 108 108 ...
 $ HarvestdateDOY : int   244 244 244 244 244 244 244 244 244 244 ...
 $ CRM           : int   114 115 116 115 116 116 116 113 115 113 ...
 $ yield         : num   12536 12456 12274 12034 11670 ...
 $ mean_maxt     : num    31 31 31 31 31 31 31 31 31 31 ...
 $ mean_mint     : num    21 21 21 21 21 21 21 21 21 21 ...
```

```

$ cum_radn      : num  3161 3161 3161 3161 3161 ...
$ cum_rain      : num  480 480 480 480 480 480 480 480 480 480 ...
$ avg_t         : num  26 26 26 26 26 26 26 26 26 26 ...
$ above_32      : int   71 71 71 71 71 71 71 71 71 71 ...
$ soil_textural_class: chr  "sharkey clay" "sharkey clay" "sharkey clay" "sharkey clay" ...
$ row_spacing   : num  0.965 0.965 0.965 0.965 0.965 ...
$ april         : num  68.6 68.6 68.6 68.6 68.6 68.6 68.6 68.6 68.6 68.6 ...
$ may           : num  238 238 238 238 238 ...
$ june          : num  24.1 24.1 24.1 24.1 24.1 24.1 24.1 24.1 24.1 24.1 ...
$ july          : num  89.9 89.9 89.9 89.9 89.9 89.9 89.9 89.9 89.9 89.9 ...
$ august        : num  1.3 1.3 1.3 1.3 1.3 1.3 1.3 1.3 1.3 1.3 ...
$ sept         : chr   "0" "0" "0" "0" ...
$ Precipitation : num  422 422 422 422 422 ...
$ week          : chr   "April 11-20" "April 11-20" "April 11-20" "April 11-20" ...
$ CRM_group     : Factor w/ 4 levels "Group_1","Group_2",...: 3 3 3 3 3 3 3 2 3 2 ...
- attr(*, "na.action")= 'omit' Named int [1:686] 1 5 7 8 13 14 15 16 17 19 ...
..- attr(*, "names")= chr [1:686] "1" "5" "7" "8" ...

```

```

mod2_loc.b <- lme(yield ~ CRM_group * SowingdateDOY,
  random = ~1 | Year/Location,
  weights = varIdent(form= ~1 | Location),
  data= variety.testing1)
sowing_seq <- sort(unique(variety.testing1$SowingdateDOY))

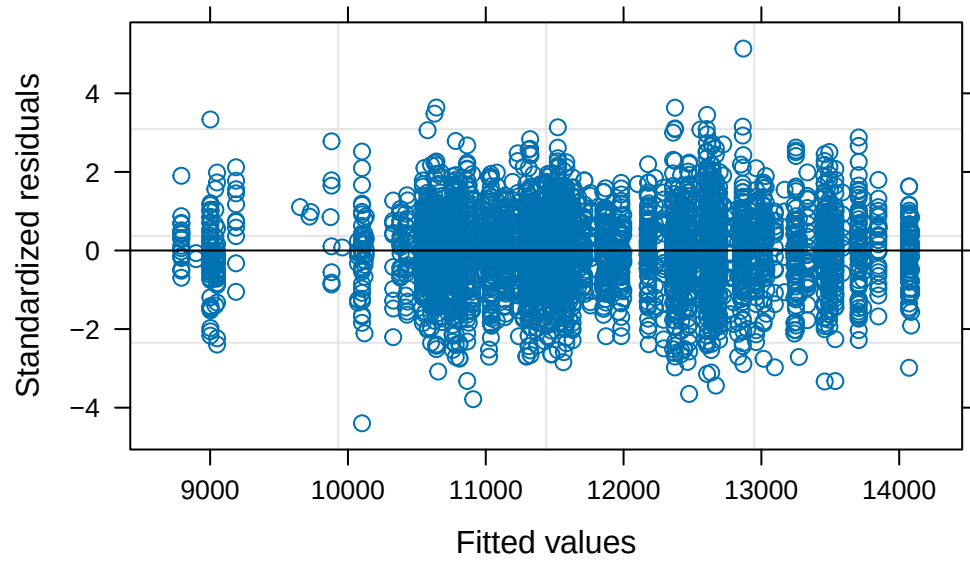
emmeans.model.b <- emmeans(mod2_loc.b, ~ SowingdateDOY,
  at = list(SowingdateDOY = sowing_seq)) %>%
  as.data.frame()

```

NOTE: Results may be misleading due to involvement in interactions

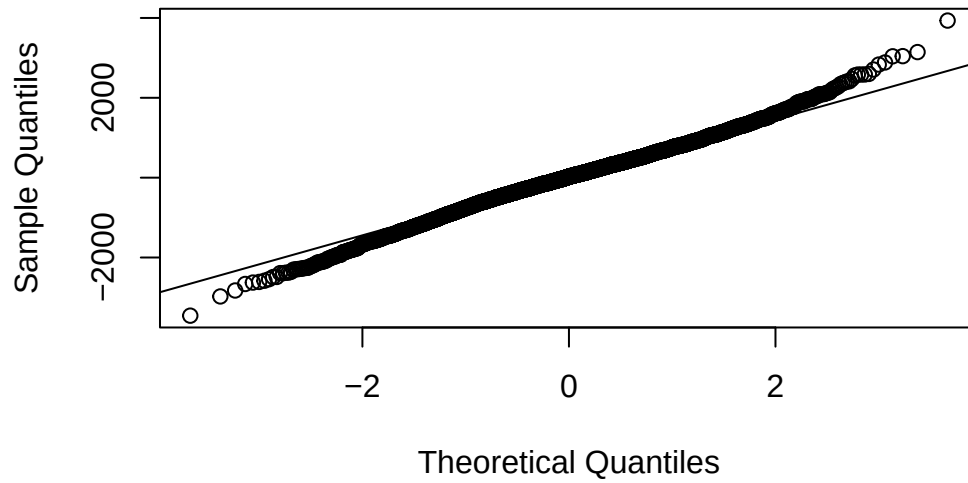
checking residuals

```
plot(mod2_loc.b)
```

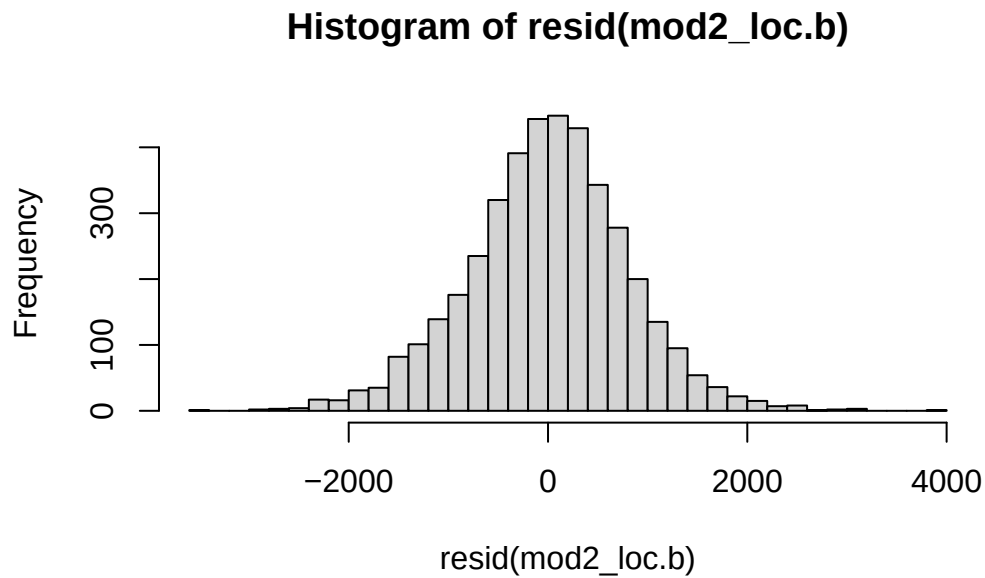


```
qqnorm(resid(mod2_loc.b)); qqline(resid(mod2_loc.b))
```

Normal Q-Q Plot



```
hist(resid(mod2_loc.b), breaks = 30)
```



```
fixed.ef <- fixef(mod2_loc.b)
intercept <- fixed.ef["(Intercept)"]
slope <- fixed.ef["SowingdateDOY"]
print(intercept)
```

```
(Intercept)
  12835.92
```

```
print(slope)
```

```
SowingdateDOY
   -19.293
```

```
em_lm <- lm(emmean ~ SowingdateDOY, data = emmeans.model.b)
summary(em_lm)
```

```
Warning in summary.lm(em_lm): essentially perfect fit: summary may be
unreliable
```

Call:

```
lm(formula = emmean ~ SowingdateDOY, data = emmeans.model.b)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-4.674e-12	-1.722e-13	1.392e-13	6.167e-13	1.119e-12

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.328e+04	1.811e-12	7.336e+15	<2e-16 ***
SowingdateDOY	-1.754e+01	1.671e-14	-1.050e+15	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

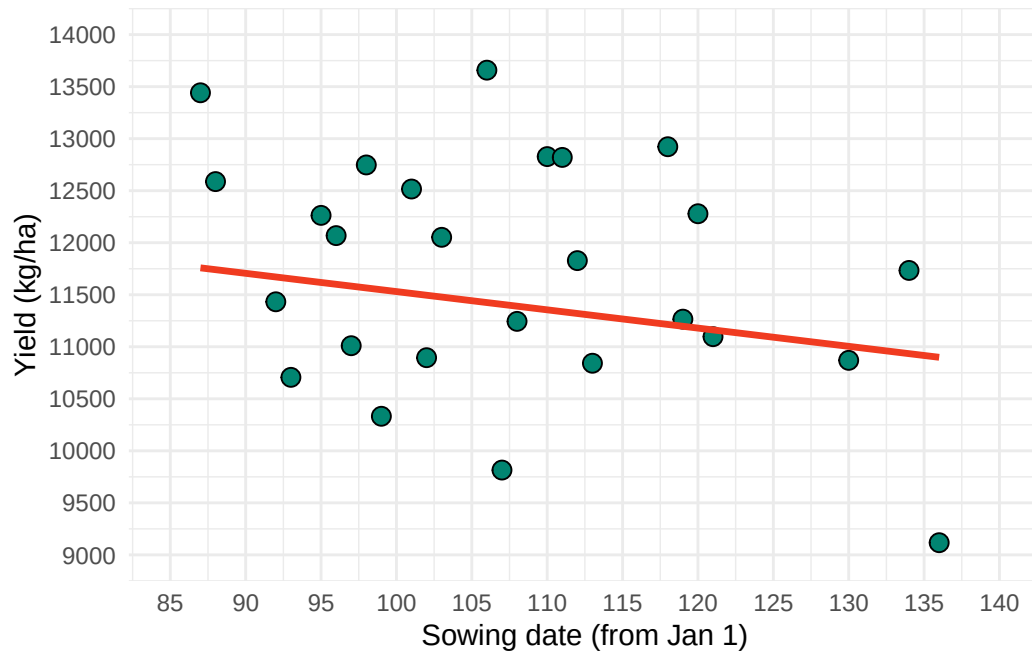
Residual standard error: 1.131e-12 on 24 degrees of freedom

Multiple R-squared: 1, Adjusted R-squared: 1

F-statistic: 1.102e+30 on 1 and 24 DF, p-value: < 2.2e-16

```
pred_emm <- data.frame(SowingdateDOY = sowing_seq)
pred_emm$yield <- predict(em_lm, newdata = pred_emm)
```

```
temp <- theme_minimal()
emmeans.model.a %>% ggplot(aes(x=SowingdateDOY,y=emmean))+
  geom_point(size=3,colour="black",fill="#018571",shape=21)+
  scale_y_continuous(limits = c(9000,14000), breaks= seq(9000, 14000, by= 500))+
  scale_x_continuous(limits = c(85,140), breaks = seq(85,140, by= 5))+
  temp+
  geom_line(data=pred_emm,
            aes(x=SowingdateDOY,y=yield),
            , linewidth = 01.2, color= "#f03b20")+
  labs(x="Sowing date (from Jan 1)",y="Yield (kg/ha)")
```



Different approach

```
library(tidyverse)
library(nlme)
library(emmeans)
library(nlraa)
```

Load data

```
variety.testing <- read.csv("2025-maximo-and-elvis/raw-data/Corn data organized-10-16-2024.csv")

variety.testing <- variety.testing %>%
  mutate(CRM_group = case_when(
    # CRM %in% c(103, 107, 108) ~ "Group_1",
    CRM %in% c(110, 111, 112, 113) ~ "Group_2",
    CRM %in% c(114, 115, 116, 117) ~ "Group_3",
    CRM %in% c(118, 119, 120, 121) ~ "Group_4"
  )) %>%
  mutate(
    Year = as.factor(Year),
    Location = as.factor(Location),
```

```

  CRM_group = as.factor(CRM_group),
  yield = as.numeric(yield),
  SowingdateDOY_f = as.factor(SowingdateDOY)
) %>%
na.omit()

```

Warning: There was 1 warning in `mutate()`.
 i In argument: `yield = as.numeric(yield)`.
 Caused by warning:
 ! NAs introduced by coercion

On this approach (mod3_loc) im using sowingdate as a factor to fit both the points and slope

```

mod3_loc <- lme(yield ~ CRM_group * SowingdateDOY_f,
  random = ~ 1 | Year/Location,
  weights = varIdent(form= ~ 1 | Location),
  data = variety.testing)

anova(mod3_loc)

```

	numDF	denDF	F-value	p-value
(Intercept)	1	3918	9737.529	<.0001
CRM_group	2	3918	25.029	<.0001
SowingdateDOY_f	25	16	2.423	0.035
CRM_group:SowingdateDOY_f	50	3918	2.372	<.0001

```
#IC_tab(mod2_loc, mod3_loc)
```

```

fixed_effects.f <- fixef(mod3_loc)
intercept <- fixed_effects.f["(Intercept)"]

```

Calculating emmeans of yield for each sowingdate.

```

overall.emmeans <- emmeans(mod3_loc, ~ SowingdateDOY_f, type = "response") %>%
  as.data.frame() %>%
  rename(SowingdateDOY = SowingdateDOY_f) %>%
  mutate(SowingdateDOY = as.numeric(as.character(SowingdateDOY)))

```

NOTE: Results may be misleading due to involvement in interactions

Fitting a linear model to obtain slope. overall slope is -27 kg/ha per day instead of -19 kg/ha per day of the previous approach. The intercept is also higher in the second approach.

```
overall.lm <- lm(emmean ~ SowingdateDOY, data = overall.emmeans)
summary(overall.lm)
```

Call:

```
lm(formula = emmean ~ SowingdateDOY, data = overall.emmeans)
```

Residuals:

Min	1Q	Median	3Q	Max
-1905.9	-711.4	134.7	773.0	1910.3

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14604.17	1714.34	8.519	1.02e-08 ***
SowingdateDOY	-26.94	15.82	-1.703	0.101

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1071 on 24 degrees of freedom

Multiple R-squared: 0.1078, Adjusted R-squared: 0.07064

F-statistic: 2.9 on 1 and 24 DF, p-value: 0.1015

Plot

```
ggplot(overall.emmeans, aes(x = SowingdateDOY, y = emmean)) +
  geom_point(size = 3, colour = "black", fill = "#018571", shape = 21) +
  #geom_line(color = "#018571", size = 1.2) +
  #geom_smooth(method= "lm")+
  geom_segment(aes(x = 86, xend = 136,
                    y = 14604.17 + (-26.94 * 86),
                    yend = 14604.17 + (-26.94 * 136)),
                color = "#f03b20", size = 1.2) +
  scale_y_continuous(limits = c(8000, 14000),
                     breaks = seq(7000,14000, by= 1000)) +
  scale_x_continuous(limits = c(85, 140), breaks = seq(80, 140, by = 5)) +
  labs(
```

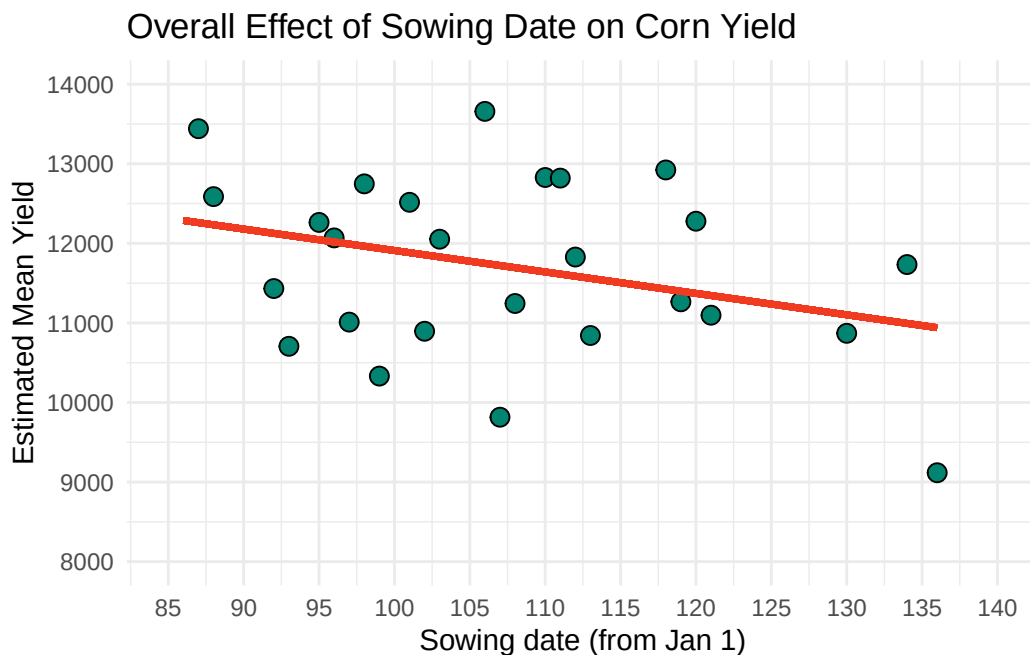
```

x = "Sowing date (from Jan 1)",
y = "Estimated Mean Yield",
title = "Overall Effect of Sowing Date on Corn Yield"
) +
theme_minimal()

```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.

Warning in geom_segment(aes(x = 86, xend = 136, y = 14604.17 + (-26.94 * : All aesthetics have a value of NA
i Please consider using `annotate()` or provide this layer with data containing a single row.



Corn Relative Maturity

```

variety.testing1$CRM <- as.factor(variety.testing1$CRM)
mod.crm <- lme(yield ~ CRM * SowingdateDOY,
               random = ~1 | Year/Location,
               weights = varIdent(form= ~1 | Location),

```

```

data= variety.testing1)
sowing_seq <- sort(unique(variety.testing1$SowingdateDOY))
summary(mod.crm)

```

Linear mixed-effects model fit by REML

Data: variety.testing1

AIC BIC logLik

66039.78 66273.05 -32982.89

Random effects:

Formula: ~1 | Year

(Intercept)

StdDev: 716.9925

Formula: ~1 | Location %in% Year

(Intercept) Residual

StdDev: 883.3421 894.0196

Variance function:

Structure: Different standard deviations per stratum

Formula: ~1 | Location

Parameter estimates:

Bellfarm Keiser Marianna Rohwer Stuttgart

1.0000000 0.9362195 0.8479410 0.8720025 0.8520032

Fixed effects: yield ~ CRM * SowingdateDOY

	Value	Std.Error	DF	t-value	p-value
(Intercept)	12760.013	3968.486	3990	3.215336	0.0013
CRM107	-435.425	4376.976	3990	-0.099481	0.9208
CRM108	1089.384	4181.094	3990	0.260550	0.7945
CRM110	-2679.568	3901.838	3990	-0.686745	0.4923
CRM111	851.481	3734.468	3990	0.228006	0.8197
CRM112	-281.243	3698.113	3990	-0.076050	0.9394
CRM113	1236.614	3685.053	3990	0.335576	0.7372
CRM114	940.958	3682.709	3990	0.255507	0.7983
CRM115	476.945	3681.362	3990	0.129557	0.8969
CRM116	1575.612	3678.009	3990	0.428387	0.6684
CRM117	614.715	3681.396	3990	0.166979	0.8674
CRM118	389.156	3686.250	3990	0.105570	0.9159
CRM119	-282.417	3746.916	3990	-0.075373	0.9399
CRM120	2718.285	3827.813	3990	0.710140	0.4777
CRM121	1105.517	4382.411	3990	0.252262	0.8009
SowingdateDOY	-26.384	35.904	40	-0.734841	0.4667

CRM107:SowingdateDOY	9.106	39.473	3990	0.230700	0.8176
CRM108:SowingdateDOY	1.530	37.609	3990	0.040690	0.9675
CRM110:SowingdateDOY	36.941	35.385	3990	1.043976	0.2966
CRM111:SowingdateDOY	6.060	33.719	3990	0.179732	0.8574
CRM112:SowingdateDOY	16.018	33.347	3990	0.480330	0.6310
CRM113:SowingdateDOY	3.680	33.219	3990	0.110769	0.9118
CRM114:SowingdateDOY	7.341	33.196	3990	0.221151	0.8250
CRM115:SowingdateDOY	11.033	33.182	3990	0.332496	0.7395
CRM116:SowingdateDOY	2.354	33.150	3990	0.070998	0.9434
CRM117:SowingdateDOY	10.644	33.183	3990	0.320774	0.7484
CRM118:SowingdateDOY	13.299	33.232	3990	0.400195	0.6890
CRM119:SowingdateDOY	17.321	33.800	3990	0.512473	0.6083
CRM120:SowingdateDOY	-8.725	34.602	3990	-0.252158	0.8009
CRM121:SowingdateDOY	16.264	40.266	3990	0.403910	0.6863

Correlation:

	(Intr)	CRM107	CRM108	CRM110	CRM111	CRM112	CRM113	CRM114
CRM107	-0.774							
CRM108	-0.808	0.736						
CRM110	-0.868	0.790	0.823					
CRM111	-0.907	0.824	0.859	0.924				
CRM112	-0.915	0.832	0.867	0.932	0.974			
CRM113	-0.918	0.834	0.870	0.936	0.977	0.986		
CRM114	-0.919	0.835	0.871	0.936	0.978	0.987	0.990	
CRM115	-0.920	0.835	0.871	0.937	0.979	0.987	0.990	0.991
CRM116	-0.920	0.836	0.872	0.938	0.979	0.988	0.991	0.992
CRM117	-0.920	0.835	0.871	0.937	0.978	0.987	0.990	0.991
CRM118	-0.918	0.833	0.869	0.935	0.977	0.986	0.989	0.990
CRM119	-0.904	0.820	0.856	0.920	0.961	0.970	0.973	0.974
CRM120	-0.886	0.804	0.838	0.902	0.942	0.950	0.953	0.954
CRM121	-0.769	0.698	0.729	0.784	0.818	0.826	0.829	0.829
SowingdateDOY	-0.994	0.767	0.801	0.861	0.899	0.907	0.910	0.911
CRM107:SowingdateDOY	0.770	-0.996	-0.732	-0.785	-0.819	-0.827	-0.830	-0.830
CRM108:SowingdateDOY	0.805	-0.734	-0.996	-0.821	-0.857	-0.865	-0.867	-0.868
CRM110:SowingdateDOY	0.859	-0.781	-0.814	-0.995	-0.914	-0.922	-0.925	-0.926
CRM111:SowingdateDOY	0.901	-0.818	-0.853	-0.918	-0.995	-0.968	-0.970	-0.971
CRM112:SowingdateDOY	0.910	-0.827	-0.863	-0.927	-0.969	-0.995	-0.980	-0.981
CRM113:SowingdateDOY	0.913	-0.830	-0.866	-0.931	-0.972	-0.981	-0.995	-0.985
CRM114:SowingdateDOY	0.914	-0.830	-0.866	-0.931	-0.973	-0.982	-0.985	-0.995
CRM115:SowingdateDOY	0.915	-0.831	-0.867	-0.932	-0.974	-0.982	-0.985	-0.986
CRM116:SowingdateDOY	0.916	-0.832	-0.867	-0.933	-0.974	-0.983	-0.986	-0.987
CRM117:SowingdateDOY	0.915	-0.831	-0.867	-0.932	-0.973	-0.982	-0.985	-0.986
CRM118:SowingdateDOY	0.913	-0.829	-0.865	-0.930	-0.972	-0.980	-0.984	-0.984
CRM119:SowingdateDOY	0.899	-0.815	-0.850	-0.915	-0.955	-0.964	-0.967	-0.968

CRM120:SowingdateDOY	0.879	-0.797	-0.832	-0.895	-0.934	-0.943	-0.946	-0.947
CRM121:SowingdateDOY	0.751	-0.681	-0.711	-0.765	-0.798	-0.806	-0.809	-0.809
	CRM115	CRM116	CRM117	CRM118	CRM119	CRM120	CRM121	SwnDOY

CRM107								
CRM108								
CRM110								
CRM111								
CRM112								
CRM113								
CRM114								
CRM115								
CRM116	0.993							
CRM117	0.992	0.993						
CRM118	0.990	0.991	0.990					
CRM119	0.975	0.975	0.975	0.973				
CRM120	0.955	0.956	0.955	0.954	0.939			
CRM121	0.830	0.830	0.830	0.829	0.815	0.798		
SowingdateDOY	0.912	0.912	0.912	0.910	0.896	0.878	0.762	
CRM107:SowingdateDOY	-0.830	-0.831	-0.831	-0.829	-0.815	-0.799	-0.694	-0.770
CRM108:SowingdateDOY	-0.868	-0.869	-0.869	-0.867	-0.853	-0.836	-0.727	-0.806
CRM110:SowingdateDOY	-0.927	-0.927	-0.926	-0.925	-0.910	-0.892	-0.775	-0.859
CRM111:SowingdateDOY	-0.972	-0.973	-0.972	-0.970	-0.955	-0.936	-0.812	-0.902
CRM112:SowingdateDOY	-0.982	-0.983	-0.982	-0.980	-0.964	-0.945	-0.821	-0.911
CRM113:SowingdateDOY	-0.985	-0.986	-0.985	-0.984	-0.968	-0.948	-0.824	-0.914
CRM114:SowingdateDOY	-0.986	-0.987	-0.986	-0.985	-0.969	-0.949	-0.825	-0.915
CRM115:SowingdateDOY	-0.995	-0.988	-0.987	-0.985	-0.970	-0.950	-0.825	-0.916
CRM116:SowingdateDOY	-0.988	-0.995	-0.988	-0.986	-0.970	-0.951	-0.826	-0.916
CRM117:SowingdateDOY	-0.987	-0.988	-0.995	-0.985	-0.970	-0.950	-0.825	-0.916
CRM118:SowingdateDOY	-0.985	-0.986	-0.985	-0.995	-0.968	-0.949	-0.824	-0.914
CRM119:SowingdateDOY	-0.969	-0.970	-0.969	-0.968	-0.995	-0.933	-0.810	-0.899
CRM120:SowingdateDOY	-0.948	-0.948	-0.948	-0.946	-0.931	-0.995	-0.792	-0.879
CRM121:SowingdateDOY	-0.809	-0.810	-0.810	-0.808	-0.795	-0.779	-0.994	-0.751
	CRM107:	CRM108:	CRM110:	CRM111:	CRM112:	CRM113:	CRM114:	

CRM107
CRM108
CRM110
CRM111
CRM112
CRM113
CRM114
CRM115
CRM116
CRM117

CRM118
 CRM119
 CRM120
 CRM121
 SowingdateDOY
 CRM107:SowingdateDOY
 CRM108:SowingdateDOY 0.737
 CRM110:SowingdateDOY 0.784 0.820
 CRM111:SowingdateDOY 0.822 0.859 0.916
 CRM112:SowingdateDOY 0.831 0.868 0.926 0.972
 CRM113:SowingdateDOY 0.833 0.871 0.929 0.975 0.985
 CRM114:SowingdateDOY 0.834 0.872 0.930 0.976 0.986 0.989
 CRM115:SowingdateDOY 0.834 0.872 0.931 0.976 0.986 0.990 0.991
 CRM116:SowingdateDOY 0.835 0.873 0.931 0.977 0.987 0.990 0.991
 CRM117:SowingdateDOY 0.834 0.873 0.931 0.976 0.986 0.990 0.991
 CRM118:SowingdateDOY 0.832 0.871 0.929 0.974 0.984 0.988 0.989
 CRM119:SowingdateDOY 0.819 0.856 0.914 0.958 0.968 0.972 0.973
 CRM120:SowingdateDOY 0.801 0.837 0.894 0.937 0.947 0.950 0.951
 CRM121:SowingdateDOY 0.683 0.716 0.763 0.800 0.809 0.812 0.813
 CRM115: CRM116: CRM117: CRM118: CRM119: CRM120:

CRM107
 CRM108
 CRM110
 CRM111
 CRM112
 CRM113
 CRM114
 CRM115
 CRM116
 CRM117
 CRM118
 CRM119
 CRM120
 CRM121

SowingdateDOY
 CRM107:SowingdateDOY
 CRM108:SowingdateDOY
 CRM110:SowingdateDOY
 CRM111:SowingdateDOY
 CRM112:SowingdateDOY
 CRM113:SowingdateDOY
 CRM114:SowingdateDOY
 CRM115:SowingdateDOY

```

CRM116:SowingdateDOY 0.992
CRM117:SowingdateDOY 0.991 0.992
CRM118:SowingdateDOY 0.990 0.990 0.990
CRM119:SowingdateDOY 0.973 0.974 0.973 0.972
CRM120:SowingdateDOY 0.952 0.953 0.952 0.950 0.935
CRM121:SowingdateDOY 0.813 0.814 0.813 0.812 0.798 0.780

```

Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-4.56055187	-0.60325702	0.02799738	0.62321545	3.87019953

Number of Observations: 4073

Number of Groups:

Year	Location	%in%	Year
14			55

```
emmeans.mod.crm <- emmeans(mod.crm,"CRM") %>% as.data.frame()
```

NOTE: Results may be misleading due to involvement in interactions

```
str(emmeans.mod.crm)
```

Classes 'summary_emm' and 'data.frame': 15 obs. of 6 variables:

```

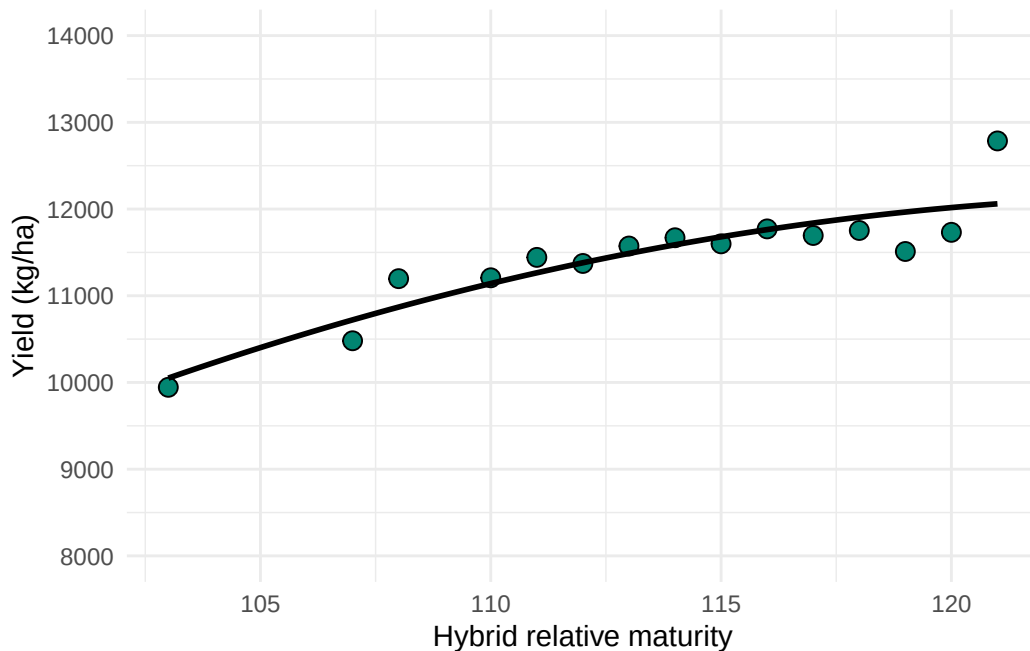
$ CRM      : Factor w/ 15 levels "103","107","108",...: 1 2 3 4 5 6 7 8 9 10 ...
$ emmean   : num  9944 10481 11197 11207 11443 ...
$ SE       : num  444 307 292 273 241 ...
$ df       : num  13 13 13 13 13 13 13 13 13 13 ...
$ lower.CL : num  8985 9819 10566 10616 10921 ...
$ upper.CL : num  10904 11143 11828 11798 11964 ...
- attr(*, "estName")= chr "emmean"
- attr(*, "clNames")= chr [1:2] "lower.CL" "upper.CL"
- attr(*, "pri.vars")= chr "CRM"
- attr(*, "adjust")= chr "none"
- attr(*, "side")= num 0
- attr(*, "delta")= num 0
- attr(*, "type")= chr "link"
- attr(*, "mesg")= chr [1:2] "Degrees-of-freedom method: containment" "Confidence level used"
- attr(*, "digits")= int 7

```

```

emmeans.mod.crm %>% ggplot(aes(x=as.numeric(as.character(CRM)),y=emmean))+
  geom_point(size=3,colour="black",fill="#018571",shape=21)+
  scale_y_continuous(limits = c(8000, 14000),
                     breaks = seq(7000,14000, by= 1000)) +
  #scale_x_continuous(limits = c(100, 125), breaks = seq(100, 125, by = 5))+
  temp+
  labs(x="Hybrid relative maturity",y="Yield (kg/ha)")+
  geom_smooth(method = "lm",formula = y ~ poly(x, 2),se=F,colour="black")

```



Slope by CRM_group

```

emmeans.crm.sw <- emmeans(mod2_loc.b, ~ SowingdateDOY|CRM_group,
                           at = list(SowingdateDOY = sowing_seq)) %>%
  as.data.frame()

```

```

sowing_slopes <- emtrends(mod2_loc.b,
                          var = "SowingdateDOY",
                          specs = ~ CRM_group)

```

```

# Convert to data.frame
sowing_slopes_df <- as.data.frame(sowing_slopes)

```

```
intercepts_df <- emmeans(mod2_loc.b, ~ CRM_group, at = list(SowingdateDOY = 0)) %>%
  as.data.frame()
```

NOTE: Results may be misleading due to involvement in interactions

```
line_data <- merge(sowing_slopes_df, intercepts_df, by = "CRM_group") %>%
  mutate(
    x_start = 85,
    x_end = 125,
    y_start = emmean + SowingdateDOY.trend * x_start,
    y_end = emmean + SowingdateDOY.trend * x_end
  )
```

```
ggplot(line_data) +
  geom_segment(aes(x = x_start, xend = x_end,
                  y = y_start, yend = y_end,
                  color = CRM_group),
              size = 1.2) +
  scale_color_manual(values = c("#d7191c", "#fdae61", "#abdda4", "#2b83ba")) +
  scale_x_continuous(limits = c(85, 140), breaks = seq(80, 140, 5)) +
  scale_y_continuous(limits = c(8000, 14000), breaks = seq(8000, 14000, 1000)) +
  labs(
    title = "Yield Response to Sowing Date by CRM Group",
    x = "Sowing Date (DOY)",
    y = "Predicted Yield (kg/ha)"
  ) +
  theme_minimal()
```

